Evaluating Limits

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Direct substitution and algebraic techniques**

Evaluate each limit:

a $\lim_{x\to 1}(3x^2-2x+5) = 6.$

b $\lim_{x\to 2}\frac{x^2-4}{x-2} = \lim_{x\to 2}(x+2) = 4.$

c $\lim_{x\to 0}\frac{\sqrt{x+4}-2}{x}$ Rationalize: $\frac{(x+4)-4}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2} \to \frac{1}{4}.$

(a) $= 6.$; (b) $= \lim_{x\to 2}(x+2) = 4.$; (c) Rationalize: $\frac{(x+4)-4}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2} \to \frac{1}{4}.$

2 **[Foundational]** • **Target: Direct substitution and algebraic techniques**

Evaluate $\lim_{x\to 3}\frac{x^2-x-6}{x^2-9}$.

Factor: $\frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{x+2}{x+3} \to \frac{5}{6}.$

3 **[Foundational]** • **Target: Trigonometric and special limits**

Using $\lim_{x\to 0}\frac{\sin x}{x} = 1$, evaluate:

a $\lim_{x\to 0}\frac{\sin 5x}{x} = 5\lim_{x\to 0}\frac{\sin 5x}{5x} = 5.$

b $\lim_{x\to 0}\frac{1-\cos x}{x} = 0$ (known result: this limit is 0, not ∞).

c $\lim_{x\to 0}\frac{\sin 3x}{\sin 5x} = \frac{3}{5}\cdot\lim\frac{\sin 3x}{3x}\cdot\frac{5x}{\sin 5x} = \frac{3}{5}.$

(a) $= 5\lim_{x\to 0}\frac{\sin 5x}{5x} = 5.$; (b) $= 0$ (known result: this limit is 0, not ∞).; (c) $= \frac{3}{5}\cdot\lim\frac{\sin 3x}{3x}\cdot\frac{5x}{\sin 5x} = \frac{3}{5}.$

4 **[Foundational]** • **Target: Trigonometric and special limits**

Evaluate $\lim_{x\to\infty}\frac{5x^3-2x+1}{2x^3+7}$.

Equal degrees: the limit is the ratio of leading coefficients, $\frac{5}{2}$.

5 **[Foundational]** • **Target: Limits that fail to exist and the squeeze theorem**

Explain why $\lim_{x\to 0}\sin\left(\frac{1}{x}\right)$ does not exist.

As $x \to 0$, $\frac{1}{x} \to \pm\infty$, so $\sin(1/x)$ oscillates between -1 and 1 infinitely often near $x = 0$ rather than approaching a single value.

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6 [Foundational] • Target: Limits that fail to exist and the squeeze theorem

Use the squeeze theorem to evaluate $\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$.

Since $-x^2 \le x^2 \cos(1/x) \le x^2$ and $\lim_{x \to 0} (\pm x^2) = 0$, the squeeze theorem gives a limit of 0.

7 [Foundational] • Target: Limits with absolute value and piecewise functions

Evaluate $\lim_{x \to 3} |x - 3| \cdot \frac{1}{x}$.

As $x \to 3$, $|x - 3| \to 0$ and $\frac{1}{x} \to \frac{1}{3}$ (a finite value), so the product limit is $0 \cdot \frac{1}{3} = 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limits with absolute value and piecewise functions

For $f(x) = 2x + 1$ if $x < 3$, and $f(x) = x^2 - 2$ if $x \ge 3$, evaluate $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$, and state whether $\lim_{x \to 3} f(x)$ exists.

$\lim_{x \to 3^-} f(x) = 2(3) + 1 = 7$. $\lim_{x \to 3^+} f(x) = 3^2 - 2 = 7$. Since both one-sided limits equal 7, $\lim_{x \to 3} f(x) = 7$ exists.

9 [Intermediate] • Target: More algebraic limit techniques

Evaluate $\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$ by factoring with the difference-of-cubes formula.

$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$, so the limit is $\lim_{x \to 2} (x^2 + 2x + 4) = 4 + 4 + 4 = 12$.

10 [Intermediate] • Target: More algebraic limit techniques

Evaluate $\lim_{h \to 0} \frac{(2 + h)^2 - 4}{h}$.

$(2 + h)^2 - 4 = 4 + 4h + h^2 - 4 = 4h + h^2 = h(4 + h)$, so the limit is $\lim_{h \to 0} (4 + h) = 4$.

11 [Intermediate] • Target: Visualizing a limit numerically and graphically

The function $f(x) = \frac{\sin x}{x}$ (undefined at $x = 0$) is graphed below. Use the graph to estimate $\lim_{x \to 0} f(x)$.

The graph approaches height 1 from both sides as $x \to 0$, even though the function has a hole exactly at $x = 0$: $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

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12 [Intermediate] • Target: Limits at infinity with different degree relationships

Evaluate each limit at infinity, based on the relative degrees of numerator and denominator:

a $\lim_{x\to\infty} \frac{4x+3}{x^2-1}$ (lower degree numerator) $= 0$ (denominator grows faster).

b $\lim_{x\to\infty} \frac{x^3-1}{2x^2+5}$ (higher degree numerator) $= \infty$ (numerator grows faster; the limit does not exist as a finite number).

(a) $= 0$ (denominator grows faster).; (b) $= \infty$ (numerator grows faster; the limit does not exist as a finite number).

13 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x\to 9} \frac{x-9}{\sqrt{x}-3}$ by rationalizing the denominator.

Multiply by $\frac{\sqrt{x}+3}{\sqrt{x}+3}$: $\frac{(x-9)(\sqrt{x}+3)}{x-9} = \sqrt{x}+3 \to \sqrt{9}+3 = 6$.

14 [Intermediate] • Target: Limits requiring conjugate multiplication

Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+3x}-x\right)$ by rationalizing.

Multiply by $\frac{\sqrt{x^2+3x}+x}{\sqrt{x^2+3x}+x}$: $\frac{3x}{\sqrt{x^2+3x}+x}$. Dividing top and bottom by x: $\frac{3}{\sqrt{1+3/x}+1} \to \frac{3}{1+1} = \frac{3}{2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 1.01 Precalculus Review for Calculus

For $f(x) = kx+1$ if $x \le 2$, and $f(x) = x^2 - 3$ if $x > 2$, find k so that $\lim_{x\to 2} f(x)$ exists.

Need $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x)$: $2k+1 = 4-3 = 1$, so $k = 0$.

16 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Evaluate $\lim_{x\to 1} \frac{x^4-1}{x-1}$ by factoring the numerator as a difference of squares twice.

$x^4-1 = (x^2-1)(x^2+1) = (x-1)(x+1)(x^2+1)$, so the limit is $\lim_{x\to 1} (x+1)(x^2+1) = (2)(2) = 4$.

17 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 11.01 Sequences and Series

Evaluate $\lim_{x\to -2} \frac{x^3+8}{x+2}$ using the sum-of-cubes factoring pattern.

$x^3+8 = (x+2)(x^2-2x+4)$, so the limit is $\lim_{x\to -2} (x^2-2x+4) = 4+4+4 = 12$.

18 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 12.01 Convergence Tests

Evaluate $\lim_{x\to 0} \frac{(3+x)^2-9}{x}$.

$(3+x)^2 - 9 = 9+6x+x^2-9 = 6x+x^2 = x(6+x)$, so the limit is $\lim_{x\to 0} (6+x) = 6$.

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19 [Challenging] • Target: More algebraic limit techniques • Cross-Lesson: 12.02 Power Series and Taylor Series

The figure shows the graph of a function $f(x)$. What is the value of $\lim_{x \to 2} f(x)$?

The limit of a function as x approaches a certain value is the y-value that the function approaches as x gets arbitrarily close to that value. Looking at the graph, as x approaches 2 from both the left and the right, the y-value of the function approaches 2. Therefore, $\lim_{x \to 2} f(x) = 2$.

20 [Challenging] • Target: Visualizing a limit numerically and graphically • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

$\frac{d}{dx}\left[(\ln x)^{\ln x}\right] =$

Let $y = (\ln x)^{\ln x} = e^{(\ln x)^2}$. $y' = e^{(\ln x)^2} \cdot 2\ln x/x = (\ln x)^{\ln x} \cdot \frac{2\ln(\ln x)}{\ln x} \cdot \frac{\ln x}{x}$... simpler: $\ln y = (\ln x)\ln(\ln x)$. $y'/y = \frac{1}{x}\ln(\ln x) + \frac{\ln x}{x\ln x} = \frac{\ln(\ln x)+1}{x}$. So $y' = (\ln x)^{\ln x} \cdot \frac{1+\ln(\ln x)}{x}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Direct Substitution and Algebraic Techniques

Evaluate $\lim_{x \to 3} \frac{x^2-9}{x^2-2x-3}$.

a. 3/2 ✓

b. 0

c. 1

d. Does not exist

Indeterminate $0/0$. Factor: $\frac{(x-3)(x+3)}{(x-3)(x+1)} = \frac{x+3}{x+1}$. Substitute $x = 3$: $\frac{3+3}{3+1} = \frac{6}{4} = \frac{3}{2}$.

22 [Foundational] • Target: Direct Substitution and Algebraic Techniques

Evaluate $\lim_{x \to 0} \frac{\sqrt{x+4}-2}{x}$.

a. 1/4 ✓

b. 1/2

c. 0

d. 1/8

Multiply numerator and denominator by conjugate $(\sqrt{x+4}+2)$: $\frac{(x+4)-4}{x(\sqrt{x+4}+2)} = \frac{1}{\sqrt{x+4}+2}$. At $x = 0$, $\frac{1}{2+2} = \frac{1}{4}$.

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23 [Foundational] • Target: Direct Substitution and Algebraic Techniques

Evaluate $\lim_{x\to 1} \frac{\frac{1}{x+1} - \frac{1}{2}}{x-1}$.

a. $-1/4$ ✓ **b.** $1/4$

c. $-1/2$ **d.** 0

Combine fractions: $\frac{2-(x+1)}{2(x+1)(x-1)} = \frac{1-x}{2(x+1)(x-1)} = \frac{-1}{2(x+1)}$. At $x=1$, $\frac{-1}{2(2)} = -\frac{1}{4}$.

24 [Foundational] • Target: Trigonometric Limits and the Squeeze Theorem

Evaluate $\lim_{x\to 0} \frac{\sin(5x)}{3x}$.

a. $5/3$ ✓ **b.** $3/5$

c. 1 **d.** 0

Rewrite as $\frac{5}{3} \cdot \frac{\sin(5x)}{5x}$. Since $\lim_{u\to 0} \frac{\sin u}{u} = 1$, the limit is $5/3$.

25 [Foundational] • Target: Trigonometric Limits and the Squeeze Theorem

Evaluate $\lim_{x\to 0} \frac{1-\cos(x)}{x^2}$.

a. $1/2$ ✓ **b.** 0

c. 1 **d.** Does not exist

Multiply by $(1+\cos x)/(1+\cos x)$: $\frac{\sin^2 x}{x^2(1+\cos x)} = \left(\frac{\sin x}{x}\right)^2 \frac{1}{1+\cos x} \to 1^2 \cdot \frac{1}{2} = \frac{1}{2}$.

26 [Foundational] • Target: Trigonometric Limits and the Squeeze Theorem

If $1 - \frac{x^2}{6} \le g(x) \le 1$ for all x near 0, find $\lim_{x\to 0} g(x)$.

a. 1 ✓ **b.** 0

c. $5/6$ **d.** Cannot be determined

Since $\lim_{x\to 0}(1 - x^2/6) = 1$ and $\lim_{x\to 0} 1 = 1$, by the Squeeze Theorem, $\lim_{x\to 0} g(x) = 1$.

27 [Foundational] • Target: Infinite Limits and Limits at Infinity

Evaluate $\lim_{x\to\infty} \frac{4x^3 - 2x + 7}{5x^3 + 3x^2 - 1}$.

a. $4/5$ ✓ **b.** 0

c. ∞ **d.** $7/(-1)$

Divide numerator and denominator by x^3: $\frac{4 - 2/x^2 + 7/x^3}{5 + 3/x - 1/x^3} \to \frac{4}{5}$ as $x \to \infty$.

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28 [Foundational] • Target: Infinite Limits and Limits at Infinity

Evaluate $\lim_{x\to-\infty} \frac{\sqrt{9x^2+4}}{2x-1}$.

a. $-3/2$ ✓

b. $3/2$

c. $9/2$

d. $-9/2$

For $x<0$, $\sqrt{x^2}=|x|=-x$. Thus $\sqrt{9x^2+4}=-x\sqrt{9+4/x^2}$. Ratio becomes $\frac{-x\sqrt{9}}{2x}=-\frac{3}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Infinite Limits and Limits at Infinity

Find $\lim_{x\to2^+} \frac{3x-1}{x-2}$ and $\lim_{x\to2^-} \frac{3x-1}{x-2}$.

a. $+\infty$ **and** $-\infty$ ✓

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 5 and 5

Numerator approaches $3(2)-1=5>0$. As $x\to2^+$, $x-2\to0^+$, so ratio $\to+\infty$. As $x\to2^-$, $x-2\to0^-$, so ratio $\to-\infty$.

30 [Intermediate] • Target: Piecewise and One-Sided Limits

Let $f(x)$ be defined as x^2+2 for $x<1$ and $4-x$ for $x\geq1$. Evaluate $\lim_{x\to1} f(x)$.

a. 3 ✓

b. 1

c. 2

d. Does not exist

Left limit: $\lim_{x\to1^-}(x^2+2)=3$. Right limit: $\lim_{x\to1^+}(4-x)=3$. Since both equal 3, the two-sided limit is 3.

31 [Intermediate] • Target: Piecewise and One-Sided Limits

Evaluate $\lim_{x\to0^+} \frac{|x|}{x}$ and $\lim_{x\to0^-} \frac{|x|}{x}$.

a. 1 **and** -1 ✓

b. 1 and 1

c. -1 and -1

d. 0 and 0

For $x>0$, $|x|=x$, so $x/x=1$. For $x<0$, $|x|=-x$, so $-x/x=-1$.

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32 [Intermediate] • Target: Piecewise and One-Sided Limits

Find the value of k such that $\lim_{x\to 2} g(x)$ exists, where $g(x)$ is $kx-3$ for $x \le 2$ and x^2+1 for $x>2$.

a. 4 ✓ **b.** 2

c. 5 **d.** 3

Require $\lim_{x\to 2^-} g(x) = \lim_{x\to 2^+} g(x) \Rightarrow 2k-3 = 2^2+1 = 5 \Rightarrow 2k = 8 \Rightarrow k = 4$.

33 [Intermediate] • Target: Infinite Limits and Limits at Infinity

One-sided limit: $f(x) = |x|/x$. $\lim_{x\to 0^-} f(x) =$

a. 1 **b.** -1 ✓

c. 0 **d.** DNE

Applying Limits and Continuity principles, For $x < 0$, $|x|/x = -1$. Therefore, the verified correct option is -1.

34 [Intermediate] • Target: Piecewise and One-Sided Limits

Evaluate the one-sided limit $\lim_{x\to -1^-} \frac{|x+1|}{x+1}$.

a. 0 **b.** ∞

c. **-1** ✓ **d.** 1

For $x \to -1^-$, $x+1$ is negative, so $\frac{|x+1|}{x+1} = -1$.

35 [Intermediate] • Target: Direct substitution and algebraic techniques

Evaluate the one-sided limit $\lim_{x\to -2^-} \frac{|x+2|}{x+2}$.

a. 0 **b.** ∞

c. 1 **d.** **-1** ✓

For $x \to -2^-$, $x+2$ is negative, so $\frac{|x+2|}{x+2} = -1$.

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36 [Intermediate] • Target: Trigonometric and special limits

Evaluate the one-sided limit $\lim_{x \to -3^-} \frac{|x+3|}{x+3}$.

a. 1

b. ∞

c. **-1** ✓

d. 0

For $x \to -3^-$, $x+3$ is negative, so $\frac{|x+3|}{x+3} = -1$.

37 [Intermediate] • Target: Limits that fail to exist and the squeeze theorem

Evaluate the one-sided limit $\lim_{x \to -4^-} \frac{|x+4|}{x+4}$.

a. 0

b. 1

c. **-1** ✓

d. ∞

For $x \to -4^-$, $x+4$ is negative, so $\frac{|x+4|}{x+4} = -1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits with absolute value and piecewise functions • Cross-Lesson: 12.01 Convergence Tests

Evaluate the one-sided limit $\lim_{x \to 3^-} \frac{|x-3|}{x-3}$.

a. 0

b. ∞

c. **-1** ✓

d. 1

For $x \to 3^-$, $x-3$ is negative, so $\frac{|x-3|}{x-3} = -1$.

39 [Challenging] • Target: More algebraic limit techniques • Cross-Lesson: 12.02 Power Series and Taylor Series

Evaluate the one-sided limit $\lim_{x \to 2^+} \frac{|x-2|}{x-2}$.

a. **1** ✓

b. -1

c. 0

d. ∞

For $x \to 2^+$, $x-2$ is positive, so $\frac{|x-2|}{x-2} = 1$.

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40 [Challenging] • Target: Visualizing a limit numerically and graphically • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Evaluate the one-sided limit $\lim_{x \to 4^+} \frac{|x-4|}{x-4}$.

a. **1** ✓

b. ∞

c. -1

d. 0

For $x \to 4^+$, $x - 4$ is positive, so $\frac{|x-4|}{x-4} = 1$.

41 [Challenging] • Target: Limits at infinity with different degree relationships • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Evaluate the one-sided limit $\lim_{x \to 5^+} \frac{|x-5|}{x-5}$.

a. 0

b. -1

c. ∞

d. **1** ✓

For $x \to 5^+$, $x - 5$ is positive, so $\frac{|x-5|}{x-5} = 1$.

42 [Challenging] • Target: Limits requiring conjugate multiplication • Cross-Lesson: 15.01 Double Integrals

Evaluate the one-sided limit $\lim_{x \to 1^+} \frac{|x-1|}{x-1}$.

a. 0

b. **1** ✓

c. ∞

d. -1

For $x \to 1^+$, $x - 1$ is positive, so $\frac{|x-1|}{x-1} = 1$.

43 [Challenging] • Target: Piecewise limits requiring a parameter • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Evaluate the one-sided limit $\lim_{x \to 6^+} \frac{|x-6|}{x-6}$.

a. **1** ✓

b. 0

c. -1

d. ∞

For $x \to 6^+$, $x - 6$ is positive, so $\frac{|x-6|}{x-6} = 1$.

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44 [Challenging] • Target: Limits involving factorable higher-degree polynomials • Cross-Lesson: 2.02 Continuity

Consider the function $f(x)$ whose graph is given below. The function is defined piecewise. For $x \le 2$, $f(x) = ax + b$. For $x > 2$, $f(x) = cx + d$. The graph of $f(x)$ passes through the point $(0, 3)$. The limit of $f(x)$ as x approaches 2 exists. Find the value of a.

a. -1 ✓

b. 1

c. 2

d. -2

For the limit of $f(x)$ as x approaches 2 to exist, the left-hand limit must equal the right-hand limit. That is, $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x)$. For $x \le 2$, $f(x) = ax + b$. So, the left-hand limit is $\lim_{x\to 2^-} (ax + b) = 2a + b$. For $x > 2$, $f(x) = cx + d$. So, the right-hand limit is $\lim_{x\to 2^+} (cx + d) = 2c + d$. Therefore, $2a + b = 2c + d$. From the graph, we can see that the function passes through $(0, 3)$. This means $f(0) = 3$. Since $0 \le 2$, we u.

45 [Challenging] • Target: Direct Substitution and Algebraic Techniques • Cross-Lesson: 3.01 The Derivative from First Principles

Squeeze theorem: $-x^2 \le f(x) \le x^2$ near $x = 0$ implies $\lim_{x\to 0} f(x) =$

a. 0 ✓

b. ± 1

c. DNE

d. 1

Applying Limits and Continuity principles, Squeeze: both bounds $\to 0$. Therefore, the verified correct option is 0.

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